UNITED INTERNATIONAL UNIVERSITY

Department of Computer Science and Engineering (CSE)

Course Syllabus

1	**Course Title**	Software Engineering Lab
2	**Course Code**	CSE 3422
3	**Trimester and Year**	Fall 2026
4	**Pre-requisites**	CSE 3412 - System Analysis and Design Lab
5	**Credit Hours**	1.0
6	**Section**	
7	**Class Hours**	
8	**Class Room**	
9	**Instructor's Name**	
10	**Email**	
11	**Office**	
12	**Counselling Hours**	Check UCAM
13	**Text Book**	Software Engineering, Eighth edition, Roger Pressman, 2015
14	**Reference**	Lecture documents
15	**Course Contents (approved by UGC)**	Laboratory work based on CSE 3421
16	**Course Outcomes (COs)**	
17	**Teaching Methods**	Lecture, laboratory exercise and reports, project development.

CO	Description
CO1	Function as an individual and as a member or leader of a team.
CO2	Design and develop a software
CO3	Learn to use modern software engineering tools.
CO4	Apply modern techniques of software engineering.
CO5	Presentation on Term Project

18 CO with Assessment Methods

CO	Assessment Method	(%)
-	Attendance	5
CO3, CO1	Class Tests	40
CO1, CO2, CO5	Project Proposal Presentation	15
CO1, CO2, CO4, CO5	Project Updates	30
CO1, CO4	Project Report	10

19 Mapping of COs and Program outcomes

CO	Statement	Bloom's Domain	Program Outcome	Knowledge Profile	Complex Problem	Engineering Activities
CO1	Function as an individual and as a member or leader of a team.	C	i Individual work and teamwork	Engineering fundamentals (K3) Specialist Knowledge (K4) Engineering Design (K5) Engineering Practice (K6)	Depth of Knowledge (P1) Range of Conflicting requirements(P2) Depth of Analysis (P3) Extent of stakeholder (P6) Interdependence (P7)	Range of Resources (A1) Level of Interaction (A2)
CO2	Design and develop a software	C	c Design/Development of solutions			
CO3	Learn to use modern software engineering tools.	C	e Modern Tool Usage			
CO4	Apply modern techniques of software engineering.	C	b Problem analysis			
CO5	Presentation on Term Project	A	J Communication			

20 Lab Outline

Lab	Topics	CO	References	Activities
1	Introduction	CO1, CO2	Lecture documents	Lecture, practice
2	CRUD Operations	CO1, CO2	Lecture documents	Lecture, practice
3	**Project Proposal Presentation**	CO1, CO2, CO5		Presentation

4	CRUD Operations + GIT	CO1, CO2, CO3	Lecture documents	Lecture, practice
5	**CT1 – CRUD Operations**	CO1, CO2		Exam
6	GIT	CO3	Lecture documents	Lecture, practice
7	**Project Update 1 (30%)**	CO1, CO2, CO4, CO5		Evaluation
8	**CT2 - GIT**	CO3		Exam
9	Testing (Selenium)	CO1, CO2	Lecture documents	Lecture, practice
10	**Project Update 2 (70%)**	CO1, CO2, CO4, CO5		Evaluation
11	**CT3 - Testing**	CO1, CO2		Exam
12	**Project Update 3 (100%)**	CO1, CO2, CO4, CO5		Evaluation

Appendix 1: Assessment Methods

Assessment Types	**Marks**
Attendance	5%
Class Test	40%
Project Proposal Presentation	15%
Project Report	10%
Project Updates	30%

Appendix 2: Grading Policy

Letter Grade	**Marks %**	**Grade Point**	**Letter Grade**	**Marks%**	**Grade Point**
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33
A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

POs	**Program Outcomes**
PO1	An ability to apply knowledge of mathematics, science, and engineering
PO2	An ability to identify, formulate, and solve complex engineering problems
PO3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
PO4	An ability to investigate complex problems using research-based knowledge and research

	methods design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
PO7	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts
PO8	An understanding of professional and ethical responsibility
PO9	An ability function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings
PO10	An ability to communicate effectively
PO11	Project management and finance
PO12	A recognition of the need for, and an ability to engage in life-long learning